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RESEARCH ARTICLE

Tool-MAD: A Multi-Agent Debate Framework for Fact Verification With Diverse Tool Augmentation and Adaptive Retrieval

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ABSTRACT Large Language Models (LLMs) often suffer from hallucinations and factual inaccuracies, particularly in complex reasoning and fact verification tasks. Multi-Agent Debate (MAD) systems aim to improve answer accuracy by enabling multiple LLM agents to engage in dialogue, promoting diverse reasoning and mutual verification. However, existing MAD frameworks primarily rely on internal knowledge or static documents, making them vulnerable to hallucinations. While multi-agent debate with knowledge-enhanced framework (MADKE) introduces external evidence to mitigate this issue, its one-time retrieval mechanism limits adaptability to new arguments or emerging information during the debate. To address these limitations, we propose Tool-MAD, a multi-agent debate framework that enhances factual verification by assigning each agent a distinct external tool, such as a search API or a RAG module. Tool-MAD introduces three key innovations: (1) a multi-agent debate framework in which agents leverage heterogeneous external tools to encourage diverse perspectives, (2) an adaptive query formulation mechanism that iteratively refines evidence retrieval as the debate progresses, and (3) the integration of Faithfulness and Answer Relevance scores into the final decision process, enabling the Judge agent to quantitatively evaluate response grounding and question alignment. Experimental results on four fact verification benchmarks show that Tool-MAD consistently outperforms MAD and achieves stronger or comparable performance than MADKE, with performance improvements of up to 11.2 percentage points over MAD and 8.0 percentage points over MADKE. Furthermore, experiments in medically specialized domains demonstrate that Tool-MAD maintains stable performance across different tool configurations and domain conditions, suggesting strong potential for broader real-world fact-checking applications.

INDEX TERMS Large language models, multi-agent debate, fact verification, tool-augmented agent.

I. INTRODUCTION

Large Language Models (LLMs) have recently demonstrated strong performance across a wide range of natural language processing tasks [1], [2], [3], including dialogue generation, summarization, and knowledge extraction. However, LLMs often suffer from hallucination, generating fluent yet factually incorrect statements that are not supported by reliable evidence [4]. This issue is particularly critical in fact verification tasks, where models must determine the correctness of a claim based on external knowledge. Fact verification also involves uncertainty-aware decision-making, because available evidence may be incomplete, conflicting, or temporally unstable. Prior studies on reliable intelligent modeling have shown that modeling uncertainty and rule-level reliability is important for trustworthy decision-making [5], [6]. These findings motivate the need for mechanisms that assess evidence reliability and decision stability in LLM-based fact verification.

Several approaches have been proposed to improve reasoning reliability. Prompt-based methods such as Chain-of-Thought (CoT) [7] and self-reflection [8] guide models

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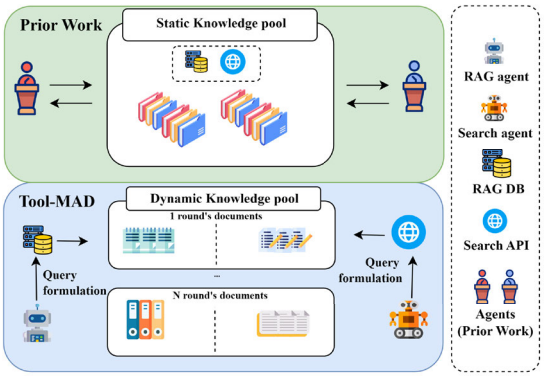


FIGURE 1. Comparison of MAD [9], MADKE [11] and the proposed Tool-MAD: Unlike MAD and MADKE, which rely on a fixed document pool for retrieval, Tool-MAD dynamically retrieves external documents at each round through query formulation.

to generate intermediate reasoning steps or revise their outputs. However, these approaches rely on a single reasoning trajectory and often fail to correct incorrect conclusions, a limitation referred to as the *Degeneration of Thought* problem [9].

To address this issue, recent studies have explored multi-agent debate frameworks in which multiple LLM agents exchange arguments and critique each other's reasoning [9], [10]. While such approaches improve reasoning robustness, most existing debate frameworks rely on either internal model knowledge or a fixed set of retrieved documents. Even when external evidence is incorporated, the retrieved documents typically remain static throughout the debate. For example, multi-agent debate with knowledge-enhanced framework (MADKE) [11] retrieves supporting documents prior to the debate, but the evidence pool remains unchanged during the discussion. This design limits the agents' ability to adapt when new claims or information gaps emerge during the debate.

In practice, fact verification rarely depends on a single, uniform source of evidence. Different information sources provide complementary strengths. Web search APIs offer broad coverage and access to up-to-date information, whereas Retrieval-Augmented Generation (RAG) modules retrieve semantically aligned documents from curated corpora such as Wikipedia. Such heterogeneity allows different agents to ground their arguments in partially complementary evidence spaces, reducing shared retrieval blind spots and improving the robustness of the debate process. Moreover, existing debate frameworks assume that evidence retrieved before the discussion is sufficient for the entire reasoning process, overlooking the dynamic nature of argumentation. In human debates, participants often refine their claims and gather additional evidence when confronted with counterarguments.

To address these limitations, we propose **Tool-MAD**, a dynamic multi-agent debate framework that integrates heterogeneous external tools with adaptive evidence retrieval. In Tool-MAD, each debate agent is equipped with a distinct retrieval tool, such as a search API or a Retrieval-Augmented Generation (RAG) module, enabling agents to access complementary knowledge sources during the debate. Unlike previous debate frameworks that rely on static evidence pools, Tool-MAD allows agents to iteratively reformulate queries and retrieve new evidence as the debate progresses. This design allows the debate process to function not only as a reasoning mechanism but also as an iterative evidence-seeking process in which emerging disagreements trigger additional retrieval and evidence acquisition. By enabling agents to continuously expand and update their evidence space, the framework improves the factual grounding of the debate and reduces hallucination risks.

In addition, Tool-MAD introduces a stability-guided evaluation mechanism based on two complementary metrics from the RAGAS framework [12]: *faithfulness*, which measures how well an agent's claim is supported by retrieved evidence, and *answer relevance*, which evaluates how directly the response addresses the original question. These metrics jointly form a *stability score*, which serves as an auxiliary signal for the Judge agent when determining the final decision.

We evaluate Tool-MAD on four fact verification benchmark datasets. Experimental results show that Tool-MAD consistently outperforms MAD [9] and achieves stronger or comparable performance than MADKE [11], with improvements of up to 11.2 and 8.0 percentage points, respectively.

The main contributions of this paper are summarized as follows:

- We propose Tool-MAD, a multi-agent debate framework that enables fact verification through heterogeneous external tools, including search APIs and Retrieval-Augmented Generation (RAG).
- We introduce an adaptive query formulation mechanism that iteratively refines evidence retrieval during the debate process.
- We incorporate a stability-guided debate mechanism using faithfulness and answer relevance to evaluate response reliability and guide final decisions.
- Extensive experiments across four fact verification datasets and two medical QA benchmarks demonstrate that Tool-MAD consistently outperforms MAD and achieves stronger or comparable performance than MADKE, with performance improvements of up to 11.2 percentage points over MAD and 8.0 percentage points over MADKE.

II. RELATED WORKS

A. MULTI-AGENT DEBATE

Multi-agent debate frameworks have emerged as a promising direction for strengthening LLM reasoning by leveraging adversarial or collaborative interactions between agents. Early theoretical foundations stem from Minsky's "society of minds" theory [13], which views intelligence as the emergent result of multiple interacting subsystems. This idea

has resurfaced in modern LLM settings, where multiple agents exchange arguments, challenge assumptions, and refine intermediate reasoning.

Du et al. [10] demonstrated that multi-round debate enables LLMs to correct each other's errors and improve logical consistency. Liang et al. [9] introduced a tit-for-tat debate structure to combat the *Degeneration of Thought* problem identified in self-reflection methods [8]. Other works have explored more complex multi-agent coordination strategies, such as majority voting, argument diversification, or structured deliberation trees. ReConcile [14] positions agents as participants in a roundtable discussion, integrating confidence-weighted consensus to avoid dominance by a single agent.

Beyond general reasoning, multi-agent frameworks have also been applied in specialized settings such as planning, safety evaluation, and code generation. For instance, debate-based self-correction mechanisms have been explored in mathematical reasoning, where agents critique intermediate steps, and in safety-alignment contexts, where disagreement is used to uncover unsafe model behaviors [10]. These efforts show that debate can create richer reasoning traces, but they still rely heavily on internal knowledge.

MADKE [11] introduced the idea of knowledge-enhanced debate by injecting externally retrieved documents before discussion; however, the evidence pool remains unchanged throughout the debate process. Similarly, debate-enhanced RAG systems retrieve supporting evidence prior to deliberation and then freeze the document set. While these systems show empirical gains, the fixed-evidence assumption limits adaptability when new arguments emerge during the debate.

Recent work has also questioned whether consensus should be the primary objective of multi-agent debate. Free-MAD proposes a consensus-free debate framework that evaluates the debate trajectory without requiring agents to converge to the same answer, aiming to reduce conformity, error propagation, and communication overhead [15]. This perspective is relevant to Tool-MAD because the Judge agent considers the full debate history and stability scores, although the current framework still uses consensus as an early stopping condition.

Another line of work emphasizes the importance of cognitive diversity in multi-agent debate. Homogeneous reasoning patterns can cause agents to remain within a fixed mental set, while assigning distinct reasoning strategies can encourage more diverse and complementary arguments [16]. Tool-MAD is complementary to this direction because it focuses on tool-level diversity by grounding agents in different evidence sources. Combining tool-level diversity with explicit reasoning-strategy diversity remains a promising direction for future work.

Beyond textual answer debate, LLM-based multi-agent systems have also been explored for numerical decision aggregation, such as weight assignment in spatial multi-criteria decision-making [17]. Although this setting differs from fact verification, it illustrates the broader use of LLM-based MAS for reliability-oriented decision support.

B. FACT VERIFICATION AND FACTUALITY EVALUATION

Ensuring the factual correctness of LLM outputs is a long-standing challenge [4], [18]. Early fact verification approaches relied on supervised classification models over structured evidence sources such as Wikipedia or news corpora. With the rise of generative models, research has shifted toward grounding generation in external corpora through retrieval-augmented generation (RAG) [19]. RAG-based fact-checkers [20] and hybrid evidence selection frameworks have shown that integrating retrieval significantly reduces hallucination in open-domain QA and claim verification. Hybrid approaches to fake news detection have also shown that linguistic cues and knowledge-based verification signals can provide complementary evidence for improving misinformation detection reliability [21]. This direction is closely related to our motivation, as Tool-MAD also aims to improve fact verification by integrating heterogeneous evidence sources within a multi-agent debate process.

Fact verification evaluation has similarly diversified. Works such as FEVER and its successors introduced label-based evaluation pipelines. More recent frameworks such as FactScore [22] attempt to quantify factuality at the claim level by decomposing model outputs. Consistency-based factuality approaches, including self-consistency [23] and re-asking [24], evaluate factual correctness by measuring the stability of model outputs rather than relying solely on external grounding.

RAGAS [12] bridged retrieval-based reasoning and factuality evaluation by introducing two complementary metrics: *faithfulness*, measuring whether claims match retrieved evidence, and *answer relevance*, measuring whether responses address the question directly. Several subsequent works have adopted faithfulness for hallucination detection in RAG systems or used relevance-based filtering to identify unusable generations. However, these metrics have predominantly been applied as post-hoc evaluators for fully generated outputs, not as real-time signals that influence multi-step reasoning.

However, prior multi-agent debate frameworks have not fully explored how factuality-oriented signals can be used as internal control signals during the debate process. Tool-MAD addresses this gap by incorporating faithfulness and answer relevance as round-level stability indicators that help assess response grounding, claim alignment, and final judge decisions. This use of factuality signals differs from existing post-hoc factuality scoring frameworks because the signals are incorporated into the debate process rather than applied only after generation.

C. TOOL-AUGMENTED AND RETRIEVAL-AUGMENTED AGENTS

A growing line of research explores augmenting LLMs with external tools, enabling them to perform tasks requiring specialized knowledge or computation. Toolformer [25] enables models to learn API-calling behaviors, while HuggingGPT [26] and GEAR [27] treat the LLM as a coordinator for heterogeneous models or systems. Tool-use frameworks have since evolved into modular agent pipelines that combine information extraction, symbolic reasoning, multi-step planning, or domain-specific simulators.

In scientific and professional domains, tool-augmented agents such as ChemCrow [28] and biomedical retrieval agents [29] demonstrate that domain-specific tools can dramatically improve reasoning fidelity. Retrieval modules, including dense retrievers, hybrid rankers, and cross-encoder rerankers, enable models to ground their reasoning in curated evidence.

Recent adaptive RAG frameworks further show that retrieval should not be treated as a fixed one-time operation. Self-RAG learns to retrieve, generate, and critique outputs through self-reflection, enabling retrieval to be invoked adaptively when needed [30]. Active retrieval methods such as FLARE similarly decide when and what to retrieve during generation, addressing the limitation of conventional retrieve-then-generate pipelines [31]. Adaptive-RAG further selects retrieval strategies according to question complexity [32]. Tool-MAD differs from these approaches by using multi-agent disagreement and debate history as triggers for query reformulation and evidence updates.

Many existing tool-augmented systems primarily adopt a single-agent perspective or use tools in a one-shot or sequential fashion. As a result, the role of multi-agent interaction in tool selection, evidence diversification, and iterative grounding remains relatively underexplored. Likewise, many conventional retrieval systems perform a single retrieval step at the beginning of a task, without adapting to evolving reasoning trajectories or emerging counterarguments. As a result, these systems often fail to revisit or refine earlier tool calls when new uncertainties arise, leading to brittle reasoning paths. Moreover, limited interaction between agents prevents the system from fully leveraging disagreement or complementary viewpoints to guide more targeted retrieval or tool use.

III. TOOL-MAD

In this section, we introduce Tool-MAD, a novel multi-agent debate framework designed to enhance the factual reliability of LLMs for claim verification tasks. Unlike previous methods that rely primarily on static evidence or single-agent reasoning, Tool-MAD incorporates iterative retrieval of external evidence and dynamic interactions among multiple specialized agents. By repeatedly updating evidence and challenging one another's conclusions, the system progressively refines its reasoning, mitigates hallucinations, and promotes more reliable consensus formation.

The framework relies on several key concepts. Adaptive query formulation refers to agents generating or refining retrieval queries based on the input claim, retrieved evidence, and debate history. The Stability Score is an internal quality-control signal that evaluates whether an agent response is grounded in the retrieved evidence and relevant to the claim. It consists of faithfulness, which measures evidential support, and answer relevance, which measures claim relevance. The agents use a shared prompt template to maintain the same task definition, output format, and debate protocol while relying on different retrieval tools and evidence sources. These components are described in the following subsections.

A. EXTERNAL TOOLS

To enable dynamic, context-aware fact verification, Tool-MAD equips agents with external retrieval tools that provide access to relevant and timely evidence during debates. Specifically, we integrate two complementary

Algorithm 1 Tool-MAD Framework

Input: claim c, debater set $\mathcal{A}$, max rounds T, stability thresholds $St = (f, ar)$
Output: final decision answer
Initialize history $H \leftarrow \emptyset$
for each $A \in \mathcal{A}$ **do**
  $S^A \leftarrow 0$
end for
for $r = 1$ **to** T **do**
  for each agent $A \in \mathcal{A}$ **do**
    Let $\bar{A}$ be the opponent of A
    if $r = 1$ **then**
      $q_A^{(r)} \leftarrow Query(A, c)$
      $D_A^{(r)} \leftarrow Retrieve(A, q_A^{(r)})$
      $a_A^{(r)} \leftarrow Respond(A, D_A^{(r)}, c)$
    else
      $q_A^{(r)} \leftarrow Query(A, c, a_{\bar{A}}^{(r-1)}, q_A^{(r-1)})$
      $D_A^{(r)} \leftarrow Retrieve(A, q_A^{(r)})$
      $a_A^{(r)} \leftarrow Respond(A, D_A^{(r)}, c, a_{\bar{A}}^{(r-1)})$
    end if
    $f_A^{(r)} \leftarrow faithfulness(a_A^{(r)}, D_A^{(r)})$
    $ar_A^{(r)} \leftarrow answerRelevance(c, a_A^{(r)})$
    $S^A \leftarrow S^A + (f_A^{(r)}, ar_A^{(r)})$
  end for
  Append round-r info to H
  if $a_R = a_S$ **and** $S^R > St$ **and** $S^S > St$ **then**
    return a_R
  end if
end for
return $Judge(A_J, H, \{S^A\}_{A \in \mathcal{A}}, c)$

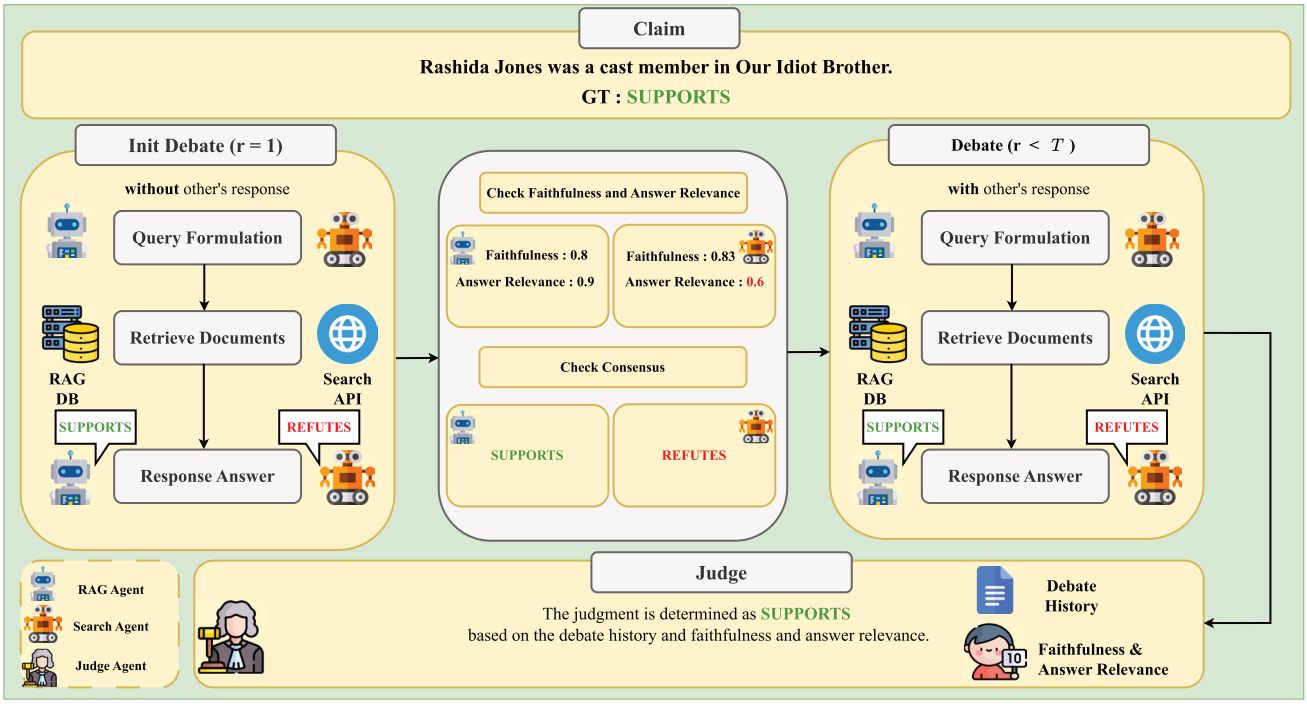


FIGURE 2. Given a claim, two agents (RAG and Search) engage in multi-round debates, where r denotes the current round and T is the predefined round threshold. If no consensus is reached or the stability score falls below the threshold by round T, the Judge Agent issues a final verdict based on the debate history and the stability score.

retrieval mechanisms: a RAG module leveraging a static corpus, and a live web Search API for real-time information access.

1) RETRIEVAL-AUGMENTED GENERATION

We employ the RAG framework [33] to augment agent reasoning with relevant documents retrieved from an embedded vector store. For this purpose, we use Milvus [34], a scalable and efficient vector database optimized for high-dimensional corpus management. We index a corpus constructed from Wikipedia articles, enabling rapid semantic retrieval. At inference time, each query returns the top three most semantically relevant documents, providing agents with targeted supporting evidence.

2) SEARCH API

Complementing the static RAG corpus, Tool-MAD also incorporates a real-time Search API, enabling agents to access up-to-date information directly from the web. For this, we utilize the Tavily Search API,[1] known for its effective integration with language models. Similar to the RAG system, the Search API retrieves the three most relevant documents per query, ensuring comprehensive coverage of evolving knowledge demands during the debate.

[1]https://tavily.com/

B. DEBATE PARTICIPANTS

1) DEBATER AGENTS

Tool-MAD involves two specialized debater agents: a RAG-based agent (A_R) that retrieves evidence from a static vector-based corpus, and a Search-based agent (A_S) that accesses live web documents via a search API. Both agents operate under a shared prompt template to ensure consistent behavior across rounds. The shared prompt template means that both agents follow the same task instruction and output format, while differing in their assigned external tool and retrieved evidence. This design controls prompt-induced variation and isolates the effect of tool heterogeneity. However, it does not explicitly impose different reasoning strategies across agents. In the first round, each agent independently generates a response based solely on the input claim. In subsequent rounds, agents refine their responses by incorporating the previous response from their opponent, enabling richer reasoning and exposure to diverse viewpoints.

2) JUDGE AGENT

If the two debater agents fail to reach consensus within a predefined maximum number of rounds, the final decision is made by a third agent, the Judge (A_J). The Judge determines the outcome based on three primary inputs: (1) the original claim, (2) the full debate history—including agent responses and retrieved evidence from each round, and (3) a stability score that assesses the factual consistency and relevance of each agent's arguments.

C. STABILITY SCORE

To quantitatively evaluate the reliability of each agent's response, Tool-MAD adopts two core metrics from the RAGAS framework [12]: *faithfulness* and *answer relevance*. These two metrics jointly form the *Stability Score* used throughout the debate.

1) FAITHFULNESS

measures how accurately an agent's response reflects the content of the retrieved evidence. Each response $a_s(q)$ is decomposed into a set of factual statements:

$$S(a_s(q)) = \{s_1, s_2, \ldots, s_{|S|}\}.$$

For each statement s_i, the LLM determines whether it can be inferred from the retrieved context $c(q)$. We define a verification function $v(s_i, c(q))$ such that $v(s_i, c(q)) = 1$ if s_i is supported by $c(q)$, and $v(s_i, c(q)) = 0$ otherwise. The faithfulness score F is then computed as the proportion of supported statements:

$$F = \frac{\sum_{i=1}^{|S|} v(s_i, c(q))}{|S|}.$$

A high faithfulness score indicates that the response is well-grounded and factually consistent with the retrieved evidence.

2) ANSWER RELEVANCE

assesses how directly the agent's response addresses the original question. For the given answer $a_s(q)$, the LLM generates a set of n potential questions:

$$Q' = \{q_1, q_2, \ldots, q_n\},$$

where each q_i is intended to represent a question that the answer could plausibly correspond to. We compute embeddings for the original question q and each generated question q_i, and calculate their cosine similarity:

$$sim(q, q_i) = \cos\left(\mathbf{e}(q), \mathbf{e}(q_i)\right).$$

The answer relevance score is then obtained by averaging the similarities:

$$AR = \frac{1}{n} \sum_{i=1}^{n} sim(q, q_i).$$

A high relevance score suggests that the answer remains focused on the intended question, whereas a low score indicates incompleteness or topic drift. It is important to note that answer relevance is not used as an external evaluation metric for fact verification labels. Instead, it functions as an internal quality-control signal that detects whether an agent's response remains aligned with the original claim during multi-round debate. This is necessary because an answer can be grounded in retrieved evidence but still drift away from the specific claim being verified.

Both metrics are computed at every debate round. If either faithfulness or answer relevance falls below a predefined threshold, the round is marked as inconclusive and the debate continues.

3) THRESHOLD OF STABILITY SCORE

To determine suitable stability-score thresholds, we analyzed the empirical distributions of faithfulness and answer relevance (Figure 3). Answer relevance was strongly skewed toward high values, with most scores above 0.8, making 0.8 a natural cutoff that filters low-quality outputs without affecting the majority of valid ones. In contrast, faithfulness exhibited a wider spread, with many responses falling between 0.7 and 0.8; thus, using a stricter threshold would unnecessarily reject reasonable, evidence-aligned answers and prolong debates.

These observations reveal a key trade-off: overly strict thresholds increase precision but hurt efficiency, especially when models produce borderline yet acceptable outputs. Therefore, we adopt 0.7 for faithfulness and 0.8 for answer relevance, which empirically provides stable convergence while maintaining decision quality.

D. TOOL-MAD PROCEDURE

Tool-MAD proceeds in multiple rounds of interaction between two debater agents, each equipped with distinct external tools. Let c denote the input claim, and $r \in \{1, \ldots, T\}$ be the current debate round, where T is the maximum allowed number of rounds. The framework involves three agents: a retrieval-based debater A_R, a search-based debater A_S, and a Judge agent A_J. Each debater generates arguments based on its retrieved evidence, and the responses in each round are quantitatively evaluated using faithfulness and answer relevance scores. If either agent's response fails to meet predefined thresholds for these scores, or if no consensus is reached, the debate proceeds to the next round. If consensus is not achieved by the final round T, the Judge agent produces a final verdict based on the accumulated dialogue and retrieved evidence. The overall framework overview is presented in Figure 2, and the complete algorithmic details are provided in the algorithm 1.

1) INITIALIZATION ROUND ($R = 1$)

In the first round, each debater independently constructs an initial query based solely on the input claim c, without reference to the opponent's argument. This query is submitted to the agent's designated retrieval tool, which returns a set of top-k relevant documents. The agent then composes its response by reasoning over the claim and the retrieved evidence. For agent $A \in \{A_R, A_S\}$, the process is formally defined as:

$$q_A^1 = Query(A, c) \tag{1}$$

$$D_A^1 = Retrieve(A, q_A^1) \tag{2}$$

$$a_A^1 = Respond(A, D_A^1, c) \tag{3}$$

where q_A^1 is the retrieval query, D_A^1 is the set of retrieved documents, and a_A^1 is the generated response for round 1.

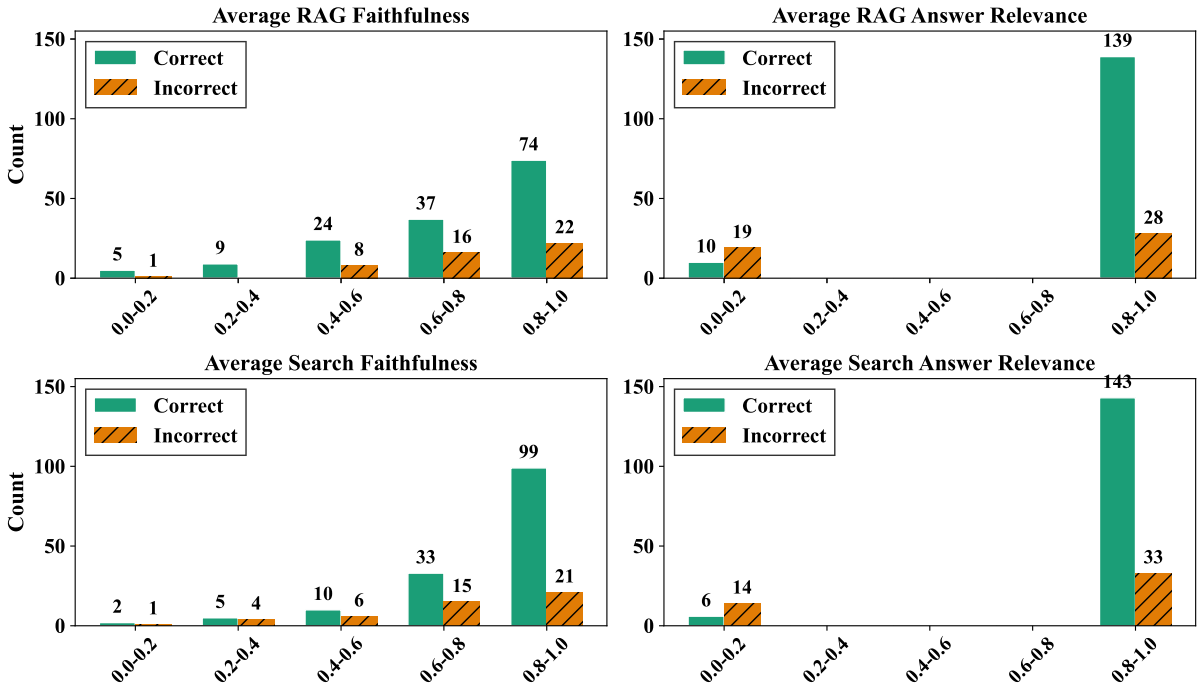


FIGURE 3. Empirical distributions of faithfulness and answer relevance for selecting stability-score thresholds. Answer relevance is concentrated above 0.8, while faithfulness shows a wider spread around 0.7–0.8. We therefore set thresholds to 0.7 (faithfulness) and 0.8 (answer relevance) to balance precision and efficiency.

2) DEBATE ROUNDS ($R > 1$)

In each subsequent round, agents refine their queries and responses by incorporating the opponent's previous answer. This enables dynamic evidence updates and encourages more robust reasoning. Each round consists of three steps: query formulation, evidence retrieval, and response generation. Specifically:

$$q_A^r = Query(A, c, a_{\bar{A}}^{r-1}) \tag{4}$$

$$D_A^r = Retrieve(A, q_A^r) \tag{5}$$

$$a_A^r = Respond(A, D_A^r, c, a_{\bar{A}}^{r-1}) \tag{6}$$

where $a_{\bar{A}}^{r-1}$ is the previous response from the opposing agent $\bar{A}$.

If both agents produce the same response in any round, the debate terminates early with a consensus, considering predefined thresholds of the stability score. Otherwise, the current round is appended to the debate history. If consensus is not reached or the responses fall below the thresholds for faithfulness or answer relevance, the debate continues until round T. The Judge agent A_J then determines the final outcome. While our experiments focus on a two-agent configuration for simplicity and controlled analysis, the proposed framework is not inherently restricted to this setting. The debate structure can be naturally extended to larger agent sets with heterogeneous tools, enabling broader evidence exploration and more diverse reasoning trajectories.

TABLE 1. Datasets used in Tool-MAD, categorized by task type and corresponding references.

Dataset	Task
FEVER [35]	Fact Verification
FEVEROUS [36]	Fact Verification
FAVIQ [37]	Fact Verification
AVeriTeC [38]	Fact Verification
MedQA [39]	Medical
PubMedQA [40]	Medical

3) JUDGE DECISION(TRIGGERED ONLY WHEN THE DEBATE REACHES THE FINAL ROUND)

If the agents fail to reach a consensus by the final round T, the Judge agent A_J makes the final decision based on the original claim c, the complete debate history H, and the stability score S. Here, S is composed of the average faithfulness and answer relevance scores across all rounds, capturing the overall quality of the agents' responses. This decision process is formalized as:

$$final\ answer = Judge(A_J, H, S, c) \tag{7}$$

The LLM-based Judge agent receives the stability score S and debate history H as part of its prompt and makes the final decision by jointly considering both.

IV. EXPERIMENTS

We evaluate the effectiveness of a Tool-MAD framework that leverages external tools in performing the fact verification task. Unless specified otherwise, we adopt a controlled

TABLE 2. Structural comparison of baseline reasoning frameworks.

Model	Reasoning	Multi-Agent	Retrieval	Query Formulation
CoT (Zero-shot)	O	X	X	X
Single-Agent (ReAct)	O	X	O	O
MAD	O	O	X	X
MADKE	O	O	O	X
Tool-MAD	O	O	O	O

evaluation protocol based on 500 randomly sampled instances per dataset for the main fact verification results. For the remaining experiments, including the ablation studies, the flexibility analysis, and the stability-score analysis, we use a 200-instance subset per dataset to reduce the computational cost of the more diagnostic analyses. The sample size used in each experiment is indicated in the corresponding tables and descriptions. Performance is measured using Exact Match (EM), where a prediction is considered correct if it exactly matches the ground-truth label.

A. IMPLEMENTATION DETAILS

We use the official RAGAS Python library version 0.3.9 to compute faithfulness and answer relevance. All LLM-based agents and RAGAS judge calls use GPT-4o-mini with temperature 0.0. The Search agent uses the Tavily Search API with `max_results=3`, while the RAG agent retrieves the top three documents from the Milvus vector store. The 500-instance main evaluation was conducted in June 2026.

B. DATASETS

We evaluate our method across a wide range of tasks to assess both factual accuracy and cross-task flexibility. For fact verification, we utilize four widely used benchmark datasets: FEVER [35] and FEVEROUS [36], which are based on Wikipedia-derived claims, as well as FAVIQ [37] and AVeriTeC [38], which include claims from real world contexts. These datasets collectively span diverse domains and claim structures, allowing for a robust assessment of factual verification capabilities. To evaluate the flexibility of our framework beyond fact verification, we additionally include medical QA datasets such as MedQA [39] and PubMedQA [40], which focus on domain-specific clinical and biomedical reasoning. A detailed summary of all datasets, including scale, domain, and task type, is provided in Table 1.

C. MODELS

We employ GPT-4o-mini [41] and Llama-3.3-70B-Instruct-Turbo [42] as backbone models in our Tool-MAD framework. We also utilize GPT-4o [41], a larger variant of GPT-4o-mini, for performance comparison. To further assess the effectiveness of the proposed framework, we conduct a performance comparison with DeepseekR1 [43], a representative reasoning based model.

1) BASELINE MODEL

We evaluate Tool-MAD by comparing it with single agent reasoning and multi-agent debate baselines. A brief summary is provided below:

- **Zero-shot CoT** [44]: Zero-shot CoT is to induce reasoning with the prompt ‘‘Let’s think step by step’’.
- **Single Agent** [45]: The single-agent baseline is based on the ReAct framework, using RAG for external knowledge retrieval. ReAct enables iterative reasoning via tool use and feedback.
- **MAD** [9]: MAD is a framework that uses an interactive ‘‘tit for tat’’ debate structure, motivated by the Degeneration of Thought observed in single agents, even when self-reflection is applied.
- **MADKE** [11]: MADKE addresses the limitations of traditional MAD methods that rely solely on internal knowledge by incorporating a static evidence pool retrieved prior to the debate.

Table 2 provides a structural comparison of these baselines, highlighting their differences across four key dimensions: intrinsic reasoning capability, multi-agent interaction, retrieval usage, and query formulation. As shown in the table, Tool-MAD is the only framework that simultaneously supports all four components by integrating structured debate, iterative retrieval, and adaptive query rewriting. In contrast, prior baselines typically cover only a subset of these capabilities, resulting in more limited overall functionality. For a fair comparison, all RAG-based baselines and Tool-MAD use the same retrieval corpus, backend, embedding model, and top-k retrieval setting. We use the Wikipedia corpus adopted in DSPy [46] as the shared retrieval source. Therefore, Tool-MAD does not benefit from a larger or different retrieval corpus. Its advantage comes from the retrieval policy: unlike static retrieval methods such as MADKE, Tool-MAD reformulates queries and updates evidence during the debate.

D. FACT VERIFICATION EXPERIMENTS

We evaluate multi-agent debate models on four fact verification benchmark datasets. Table 3 presents the experimental results. Tool-MAD, using the lightweight GPT-4o-mini as its backbone, outperforms the more powerful GPT-4o across all evaluated benchmark datasets. Notably, it shows a 12.2 percentage-point improvement on AVeriTeC, the dataset with the highest label complexity, demonstrating the robustness of our framework under more challenging

TABLE 3. Main results on four fact verification datasets (FEVER, FEVEROUS, FAVIQ, and AVeriTeC). Tool-MAD consistently improves performance across both proprietary (e.g., GPT-4o, DeepSeekR1) and open-source (e.g., LLaMA-3.3-70B) models, outperforming other multi-agent debate frameworks. Average scores are reported for overall comparison. Bold indicates the highest Exact Match score within each base model group (above: GPT-4 variants and DeepSeekR1; below: LLaMA).

Model	FEVER	FEVEROUS	FAVIQ	AVeriTeC	Average
DeepSeekR1	64.4	52.6	71.2	**57.6**	61.5
GPT-4o	64.2	30.6	73.4	44.8	53.3
GPT-4o-mini	61.2	30.4	63.8	43.6	49.8
+ CoT(Zero-shot)	62.0	27.4	70.8	42.2	50.6
+ Single Agent(ReAct)	58.6	32.2	62.8	20.8	43.6
+ MAD	61.8	61.0	63.0	44.8	57.7
+ MADKE	65.8	63.6	73.8	54.6	64.5
+ Tool-MAD	**67.8**	**71.6**	**74.2**	57.0	**67.7**
Llama-3.3-70B(Inst)	67.8	32.8	71.4	43.0	53.8
+ CoT(Zero-shot)	68.6	28.4	71.2	40.4	52.2
+ Single Agent(ReAct)	54.2	32.2	68.0	34.4	52.1
+ MAD	60.2	40.2	58.4	40.8	49.9
+ MADKE	58.4	53.4	72.0	48.4	58.1
+ Tool-MAD	**69.2**	**77.4**	**77.4**	**52.6**	**69.2**

TABLE 4. Accuracy and 95% bootstrap confidence intervals across datasets. (Backbone model: GPT-4o-mini).

Dataset	Acc.	95% CI
FEVER	67.8	[63.6, 71.8]
FEVEROUS	71.6	[67.6, 75.4]
FAVIQ	74.2	[70.2, 78.0]
AVeriTeC	57.0	[52.6, 61.2]

TABLE 5. Exact Match (EM) scores of different multi-agent debate frameworks on MedQA and PubMedQA. Bold indicates the highest score.

Model	MedQA	PubMedQA
MAD	58.0	22.5
MADKE	74.0	21.5
Tool-MAD	**77.0**	**29.0**

verification settings. Compared to DeepSeekR1, Tool-MAD achieves higher accuracy on three of the four tasks, with improvements of up to 19.0 percentage points on FEVEROUS, while remaining comparable on AVeriTeC. Tool-MAD achieves comparable performance with the open-source Llama-3.3-70B backbone, confirming its robustness across model backbones. Tool-MAD outperforms other multi-agent debate frameworks, achieving up to 11.2 and 8.0 percentage-point improvements over MAD and MADKE, respectively, with average gains of 10.0 and 3.2 percentage points. These results demonstrate that Tool-MAD delivers superior performance in fact verification tasks compared to existing multi-agent frameworks, and further highlight its architectural advantage over MADKE, which also leverages external knowledge. To better understand the observed performance gap, we analyze the limitations of each baseline system. The Single Agent, despite leveraging the ReAct framework [45] and external tools, frequently generates incorrect answers due to the retrieval of irrelevant or incomplete documents. MAD [9] improves upon this through multi-agent debate, but still struggles when evidence is insufficient, often defaulting to a generic "Not Enough Info" response. MADKE [11] achieves relatively stable performance by relying on a static evidence pool retrieved prior to the debate, yet fails to generalize well on more complex datasets such as AVeriTeC and FEVEROUS, where dynamic adaptation is required. These results highlight the architectural advantage of Tool-MAD, which enables agents to iteratively retrieve new and contextually relevant evidence throughout the debate, leading to improved factual reasoning. Additionally, we report 95% bootstrap confidence intervals for Tool-MAD using GPT-4o-mini as the backbone to assess the statistical robustness of the results. Bootstrap resampling provides an estimate of performance variability without requiring repeated full-scale experiments, making it suitable for evaluating computationally intensive multi-agent debate frameworks. The resulting confidence intervals are presented in Table 4.

E. FLEXIBILITY

This experiment examines whether Tool-MAD can be adapted to domains and retrieval tools that differ from the Wikipedia-based fact verification setting. We view this experiment as a preliminary domain-transfer analysis rather than a comprehensive validation in the medical domain. For comparison, we use the debate frameworks that showed competitive performance in the fact verification experiments, excluding single-agent baselines that demonstrated relatively lower performance.

We evaluate the framework on two medical QA datasets: MedQA, which consists of clinical multiple-choice questions, and PubMedQA, which focuses on biomedical fact verification. To examine tool extensibility, we modify the retrieval components used in Tool-MAD. Specifically, the MedQA experiment uses a PubMed-based RAG corpus, while the PubMedQA setting replaces the standard web search API with OpenAlex [47], a large-scale open scholarly database.

In the PubMedQA pipeline, the agent extracts keywords from each query and retrieves three relevant abstracts from

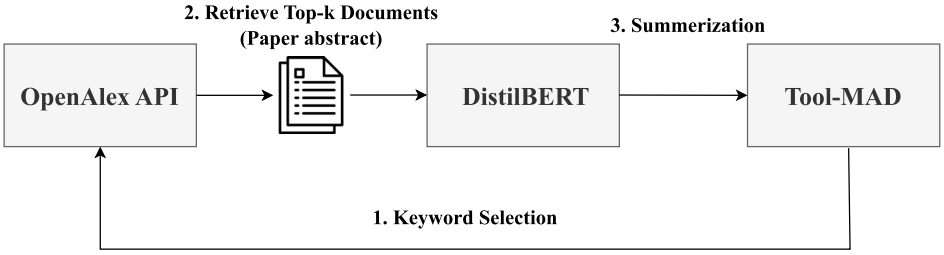


FIGURE 4. Workflow diagram of the paper retrieval API used in the PubMedQA experiment.

TABLE 6. Performance comparison of different agent combinations on FEVER, FEVEROUS, FAVIQ and AVeriTeC datasets. Tool-MAD achieves the best overall results across combinations of RAG, Search, and Vanilla agents. Bold indicates the highest Exact Match score.

Model	FEVER	FEVEROUS	FAVIQ	AVeriTeC
Vanilla + Vanilla	62.0	40.0	60.0	45.0
RAG + Vanilla	65.5	57.5	63.5	58.0
Search + Vanilla	60.5	43.0	63.5	53.5
RAG + RAG	67.5	60.5	68.5	56.5
Search + Search	67.0	60.5	68.5	55.5
Tool-MAD (RAG + Search)	**73.0**	**71.5**	**77.5**	**62.0**

OpenAlex. The retrieved abstracts are then processed using a BERT-based sentence selection module [48], which identifies the most relevant sentences for the given query. The selected evidence is incorporated into the debate process. The overall PubMedQA retrieval workflow is illustrated in Figure 4.

As shown in Table 5, Tool-MAD shows the strongest performance among the compared debate frameworks on both datasets. Notably, although MADKE incorporates external knowledge, it performs worse than MAD on PubMedQA, whereas Tool-MAD remains competitive when the underlying retrieval tools are modified. These results suggest that the proposed framework can be transferred to specialized retrieval environments, although further validation is needed before drawing broad conclusions about medical-domain reliability.

One possible explanation for this improvement is the complementary nature of the heterogeneous retrieval sources used in Tool-MAD. Medical QA and biomedical verification often require evidence from different types of sources, including structured biomedical knowledge and recently published scholarly information. Search-based retrieval can help capture external or updated evidence, while RAG-based retrieval provides semantically aligned contextual knowledge from a curated corpus. By combining these sources within the debate process, Tool-MAD enables agents to access more diverse evidence than systems that rely on a single retrieval modality.

Another observation is that Tool-MAD maintains relatively robust performance despite the domain shift from Wikipedia-based fact verification to biomedical question answering. This may be partly attributed to the framework's iterative retrieval mechanism, which allows agents to refine queries and update evidence as the debate progresses. Such iterative retrieval can help reduce errors caused by domain-specific terminology or retrieval noise.

These results should be viewed as an initial domain-transfer analysis rather than a full medical-domain validation. The experiment shows that Tool-MAD can retain its retrieval-and-debate mechanism when the underlying evidence sources are changed from Wikipedia and web search to biomedical retrieval resources. This suggests that heterogeneous tool augmentation and iterative evidence updating may support more flexible reasoning across specialized domains. Further evaluation with larger medical benchmarks and expert-validated evidence remains an important direction for future work.

F. ABLATION STUDY

1) COMBINATION OF AGENTS

To evaluate the impact of different agent configurations, we compare their performance across the FEVER, FEVEROUS, and FAVIQ benchmark datasets. All agents follow the same Tool-MAD procedure, with the only variation being the external tools they employ. Specifically, we define three types of agents: a base agent without any external tools (Vanilla), an agent using retrieval-augmented generation (RAG), and an agent equipped with a web-based search API (Search).

The results are presented in Table 6. As shown in the table, configurations that incorporate external tools consistently outperform those without tools, indicating that tool integration improves performance in fact verification tasks. Among all agent combinations tested, the original Tool-MAD setup (RAG + Search) achieved the highest average accuracy, demonstrating its robustness and effectiveness.

In particular, configurations like RAG+RAG or Search+Search outperformed the Vanilla setup, but still fell short of the hybrid configuration (RAG+Search). This suggests that relying on a single type of tool can lead to redundant retrievals

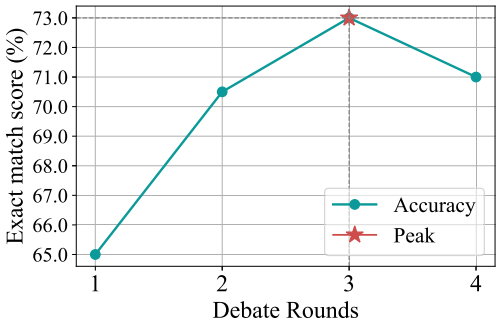


FIGURE 5. Exact match performance of Tool-MAD on the FEVER dataset across different debate rounds.

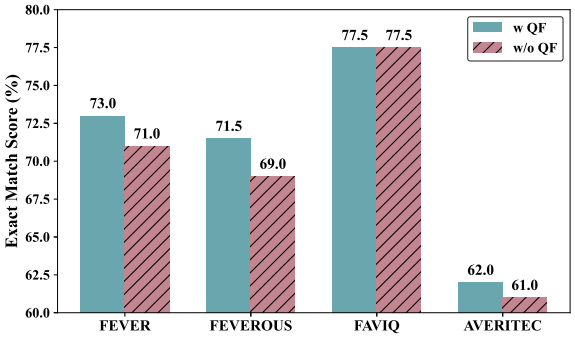


FIGURE 6. Exact Match performance comparison with and without query formulation (QF) across four benchmark datasets.

and limit the diversity of evidence, thereby constraining overall performance improvements.

To further interpret these results, we note that the hybrid configuration benefits from the complementary characteristics of the two external tools. RAG-based retrieval provides semantically aligned, context-rich documents, while search-based retrieval offers broader coverage and access to rapidly updated information. These heterogeneous evidence sources allow the agents to explore distinct perspectives during the debate, reducing shared blind spots and producing more reliable conclusions. In contrast, homogeneous configurations tend to retrieve overlapping content, which limits the diversity of arguments and constrains potential performance gains.

2) NUMBER OF DEBATE ROUNDS

We investigate how the number of debate rounds affects the overall performance of Tool-MAD. To analyze the effect of debate round, we conduct an experiment on the FEVER dataset. The results are shown in Figure 5. We observe that performance is lowest when only the Init Round (Round 1) is used. Accuracy gradually improves up to round 3 but slightly declines at round 4. Based on this observation, we set the threshold for the number of debate rounds to 3 in order to balance performance and efficiency.

This pattern suggests that although additional rounds help refine arguments and retrieve more relevant evidence, excessive debate depth may introduce unnecessary speculation or amplify intermediate reasoning errors. Such behavior aligns with observations in prior multi-step reasoning research, where prolonged interactions can lead to unstable or redundant argumentation. Therefore, three rounds provide sufficient opportunity for evidence refinement without causing degradation in the quality of the debate.

3) EFFECT OF QUERY FORMULATION

We conducted ablation experiments to evaluate the effectiveness of dynamic query formulation. In Tool-MAD, each agent can independently revise its query at every debate round to retrieve more relevant documents. To isolate the impact of this mechanism, we compare the standard Tool-MAD setup with dynamic query updates at each round (w/ Query Formulation) to a variant that uses only the initial claim for retrieval throughout all rounds (w/o Query Formulation).

As shown in Figure 6, dynamic query formulation improves performance on all datasets except FAVIQ, where the score remains unchanged. The most notable improvement is observed on FEVER (+2.0) and FEVEROUS (+2.5), followed by a modest gain on AVeriTeC (+1.0). These results demonstrate that formulating queries in response to the opponent's argument helps uncover more relevant evidence and contributes to improved fact verification accuracy.

One possible explanation for this behavior is the difference in dataset characteristics. FEVER and FEVEROUS primarily contain entity-centric factual claims where emphasizing specific entities or relations during query reformulation leads to more targeted retrieval. In contrast, FAVIQ includes broader information-seeking questions that often require diverse or multi-hop evidence, meaning that the initial query already captures most of the relevant search space. As a result, further refinement has limited impact on FAVIQ, whereas datasets with more structured claim formulations benefit more from adaptive querying.

4) EFFECT OF STABILITY SCORE

We conducted ablation experiments to assess the effectiveness of incorporating faithfulness and answer relevance (stability score) as dynamic evaluation signals. In Tool-MAD, each agent's response is evaluated at every debate round using these two metrics, and if either score falls below predefined thresholds, the round is considered inconclusive and is repeated. This mechanism enables the framework to dynamically filter out low-quality responses throughout the debate.

To isolate the impact of this score-based filtering, we compare the standard Tool-MAD configuration that utilizes faithfulness and answer relevance scores to guide debate progression (w/ Scoring Feedback) with a variant that skips score-based validation and progresses through all rounds regardless of response quality (w/o Scoring Feedback).

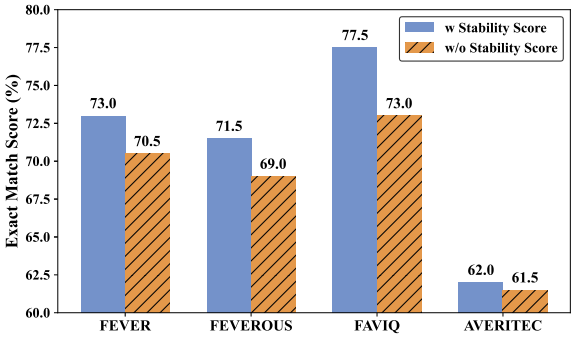


FIGURE 7. Faithfulness and Answer Relevance scores across four datasets, based on the final predictions used for Exact Match evaluation, comparing models with and without scoring-based feedback.

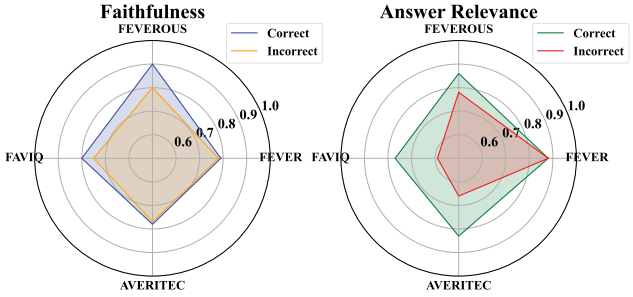


FIGURE 8. The figure presents the average Faithfulness and Answer Relevance scores for both correct and incorrect answers across four datasets.

The results in Figure 7 highlight that scoring feedback consistently improves EM performance across all datasets, with the most significant gain observed on FAVIQ (+4.5%). These findings suggest that faithfulness and answer relevance function not merely as evaluation metrics but as crucial control signals that help maintain factual consistency and question relevance throughout multi-round debates.

A notable aspect of this trend is that FAVIQ benefits particularly strongly from stability-based filtering. Because many FAVIQ questions are open-ended and sensitive to topical alignment, answers that deviate slightly from the intended question are often incorrect even when they contain factually valid information. In such cases, answer relevance becomes a highly predictive indicator of correctness, while faithfulness prevents unsupported reasoning from propagating across rounds. Together, these signals help stabilize the debate process and reduce both topic drift and hallucinated content.

G. ANALYSIS OF STABILITY SCORE

Figure 8 compares the average Faithfulness and Answer Relevance scores across four datasets, based on whether the agent's prediction was correct or incorrect. In all datasets, responses classified as correct consistently exhibit higher scores on both metrics, indicating a strong correlation between these scores and the Exact Match (EM) accuracy. These findings suggest that the scores can be used not only as post-hoc evaluation metrics but also as internal signals for supporting response filtering and final decision-making.

In particular, the FAVIQ and AVeriTeC datasets show a pronounced gap in Answer Relevance scores between correct and incorrect responses, suggesting that alignment with the original user query plays a crucial role in predicting correctness for complex question answering tasks. These results highlight that Faithfulness and Answer Relevance serve as strong and reliable indicators for evaluating response quality and factual consistency. They can also contribute to quality control and final decision-making mechanisms in multi-agent debate systems.

V. DISCUSSION

A. STRUCTURAL EFFECTS OF QUERY REWRITING

One of the most salient patterns observed in our analysis is that the gradual query rewriting performed throughout the debate plays a crucial role in improving verification accuracy. Initial queries typically take a broad, holistic form that mirrors the surface structure of the claim, but as the debate progresses, the agents refine their queries by decomposing attributes, restructuring entity relations, or explicitly specifying missing details. For instance, in the Trinidad and Tobago Guardian ownership case, the first-round query asked about the newspaper's overall ''ownership history,'' which retrieved only partial information. In later rounds, the agents rewrote the query to isolate the key relation: ''the relationship between Guardian Media Limited and ANSA McAL,'' which surfaced documents describing the hierarchical ownership chain. This shift transformed an initially ambiguous or contradictory prediction into a stable SUPPORTS decision. A similar phenomenon occurred in the Mel Ott RBI case. Early queries retrieved only season-level statistics, leading to NOT ENOUGH INFO (NEI), but rewriting the query into ''total career RBIs'' enabled retrieval of the exact aggregate value (1860), allowing the system to recover from partial evidence and converge on the correct label. These examples demonstrate that the multi-round query rewriting process is not merely repeated retrieval, but a structured mechanism that incrementally decomposes and reconstructs the claim, guiding the system toward increasingly precise evidence.

B. STRUCTURAL EFFECTS OF STABILITY SCORES

A second notable characteristic identified in the analysis is the stabilizing effect of round-wise Stability scores, which evaluate each answer's faithfulness (document–answer consistency) and answer relevance (question–answer consistency). It is common for the RAG-based answer and the Search-based answer to propose different labels for the same claim. Rather than selecting based on majority or repetition, the system computes stability scores for both answers at every round and tends to choose the label whose supporting answer

maintains consistently higher semantic alignment with the retrieved evidence and the original question. For example, in the Stomatochaeta classification case, the RAG answer repeatedly predicted SUPPORTS while the Search answer predicted REFUTES. Across multiple rounds, however, the Search answer exhibited higher faithfulness scores and more stable question–answer relevance, prompting the model to adopt REFUTES as the final label. The Obispo first ascent case further illustrates this effect. Although the first round produced NEI due to insufficient evidence, subsequent rounds produced SUPPORTS answers with steadily increasing stability scores, eventually overriding the initial uncertainty and yielding a confident final decision. Because stability score directly measures whether an answer is grounded in retrieved evidence, it remains robust even under noisy retrieval conditions, assigning greater weight to evidence-aligned answers that consistently satisfy semantic constraints. Thus, stability score provides a meta-evaluative layer that re-assesses each candidate answer based on evidence quality and coherence, substantially enhancing convergence reliability compared to single-agent RAG systems.

C. MULTI-AGENT DEBATE AS A STRUCTURAL EXPLORATION MECHANISM

Finally, the analysis shows that the debate process exhibits an emergent capacity to explore structural properties of evidence, temporal shifts, and conflicting information even without an explicit moderator. A representative example is the Fort Myers Police Department case, where the Search tool retrieved up-to-date information indicating that the police chief had recently passed away, implying REFUTES, while the ground-truth label, anchored to an earlier timestamp, remained SUPPORTS. In such scenarios, the agents did not merely default to the labeled answer but instead engaged in a comparative evaluation of evidence recency, reliability, and contextual relevance. In other cases, the agents grounded their arguments in different retrieval sources and iteratively critiqued one another's evidence, adjusting their interpretations in later rounds. This behavior suggests that the system is not simply producing parallel answers but is performing a form of quasi-critical reasoning that assesses the coherence, conflict, and incompleteness of available evidence. Such patterns indicate that multi-agent debate has the potential to evolve beyond static fact verification toward a richer inference system capable of handling temporal drift, evidence conflicts, and multi-faceted claims, conditions that more closely resemble real-world reasoning tasks.

VI. LIMITATIONS

Tool-MAD depends on the quality and availability of external retrieval sources. Retrieved evidence may be incomplete, noisy, biased, or temporally inconsistent with benchmark labels, especially when live search results change over time. The framework also increases computational cost and latency because each debate round involves query formulation, retrieval, response generation, stability-score computation, and judge-based aggregation for multiple agents. In our FEVER cost analysis, Tool-MAD required 9.45 LLM calls, 11,436 input tokens, and 2,949 output tokens per query on average, with an average latency of 51.1 seconds. A variant without the RAGAS-based Stability Score reduced the cost to 5.67 LLM calls, 5,725 input tokens, and 345 output tokens, indicating that stability-score computation is a major source of additional overhead. Although Tool-MAD improves verification accuracy, real-time deployment requires cost-aware stopping strategies, selective scoring, and efficient caching. The current implementation uses two debater agents with shared task instructions. This design isolates the effect of tool heterogeneity, but it may limit cognitive diversity. Future work could combine tool-level diversity with strategy-level diversity or consensus-free trajectory evaluation. Finally, our experiments focus mainly on text-based fact verification. Extending Tool-MAD to multimodal verification would require modality-specific tools such as visual retrieval, OCR, and image-grounding models.

VII. CONCLUSION

In this work, we introduced Tool-MAD, a multi-agent debate framework that enhances factual verification by assigning distinct external tools to each agent and enabling adaptive, context-aware retrieval throughout the debate. Across diverse fact-verification and medical QA benchmarks, Tool-MAD achieved performance improvements of up to 11.2 percentage points over MAD and 8.0 percentage points over MADKE, with ablation studies confirming the importance of tool diversity, dynamic query formulation, and our Stability Score, which functions as an internal control signal rather than a retrospective metric. These results demonstrate that combining structured debate with adaptive tool use significantly improves factual grounding and robustness. Tool-MAD's stable performance across domains and retrieval configurations indicates strong potential for extension to richer tool ecosystems and more advanced judge models. Overall, Tool-MAD provides a unified and evidence-sensitive framework for transparent and reliable multi-agent reasoning in high-stakes factual verification tasks.

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